

**Table I.** Mean predictive performance of the Logistic Regression model with one independent variable

| $n$ | Handling<br>Technique | IR    | MLE           |               |               |               | Score         |               |               |               |
|-----|-----------------------|-------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|     |                       |       | Acc           | Sens          | Spec          | BA            | Acc           | Sens          | Spec          | BA            |
| 200 | Original              | 90:10 | 0.9192        | 0.1939        | 0.9317        | 0.5969        | <b>0.9279</b> | <b>0.4561</b> | <b>0.9802</b> | <b>0.7182</b> |
|     |                       | 80:20 | 0.8837        | 0.5436        | 0.9687        | 0.7561        | <b>0.8964</b> | <b>0.6733</b> | <b>0.9522</b> | <b>0.8128</b> |
|     |                       | 70:30 | 0.8674        | 0.7621        | 0.9126        | 0.8373        | <b>0.8675</b> | <b>0.7614</b> | <b>0.9130</b> | <b>0.8372</b> |
|     | RUS                   | 90:10 | 0.8240        | <b>0.9018</b> | 0.8153        | 0.8586        | <b>0.8425</b> | 0.8852        | <b>0.8378</b> | <b>0.8615</b> |
|     |                       | 80:20 | 0.8460        | <b>0.9081</b> | 0.8305        | 0.8693        | <b>0.8622</b> | 0.8814        | <b>0.8574</b> | <b>0.8694</b> |
|     |                       | 70:30 | 0.8492        | <b>0.8917</b> | 0.8309        | 0.8613        | <b>0.8592</b> | 0.8717        | <b>0.8538</b> | <b>0.8628</b> |
|     | ROS                   | 90:10 | 0.8291        | <b>0.9003</b> | 0.8212        | 0.8608        | <b>0.8528</b> | 0.8787        | <b>0.8499</b> | <b>0.8643</b> |
|     |                       | 80:20 | 0.8484        | <b>0.9103</b> | 0.8329        | 0.8716        | <b>0.8658</b> | 0.8837        | <b>0.8613</b> | <b>0.8725</b> |
|     |                       | 70:30 | 0.8500        | <b>0.8946</b> | 0.8308        | <b>0.8627</b> | <b>0.8577</b> | 0.8724        | <b>0.8513</b> | 0.8619        |
|     | SMOTE                 | 90:10 | 0.8387        | <b>0.8960</b> | 0.8323        | <b>0.8641</b> | <b>0.8707</b> | 0.8555        | <b>0.8724</b> | <b>0.8640</b> |
|     |                       | 80:20 | 0.8495        | <b>0.9103</b> | 0.8343        | 0.8723        | <b>0.8698</b> | 0.8808        | <b>0.8671</b> | <b>0.8740</b> |
|     |                       | 70:30 | 0.8588        | <b>0.8834</b> | 0.8482        | <b>0.8658</b> | <b>0.8630</b> | 0.8579        | <b>0.8652</b> | 0.8616        |
|     | ANS                   | 90:10 | 0.8371        | <b>0.8952</b> | 0.8306        | 0.8629        | <b>0.8651</b> | 0.8633        | <b>0.8653</b> | <b>0.8643</b> |
|     |                       | 80:20 | 0.8495        | <b>0.9090</b> | 0.8346        | 0.8718        | <b>0.8696</b> | 0.8809        | <b>0.8667</b> | <b>0.8738</b> |
|     |                       | 70:30 | 0.8585        | <b>0.8816</b> | 0.8486        | <b>0.8651</b> | <b>0.8636</b> | 0.8576        | <b>0.8662</b> | 0.8619        |
|     | DBSMOTE               | 90:10 | 0.8430        | <b>0.8863</b> | 0.8381        | <b>0.8622</b> | <b>0.8856</b> | 0.8182        | <b>0.8931</b> | 0.8556        |
|     |                       | 80:20 | 0.8538        | <b>0.8981</b> | 0.8428        | <b>0.8704</b> | <b>0.8810</b> | 0.8502        | <b>0.8886</b> | 0.8694        |
|     |                       | 70:30 | 0.8592        | <b>0.8754</b> | 0.8522        | <b>0.8638</b> | <b>0.8654</b> | 0.8383        | <b>0.8770</b> | 0.8576        |
| 500 | Original              | 90:10 | 0.9068        | 0.0821        | 0.9497        | 0.5403        | <b>0.9245</b> | <b>0.4487</b> | <b>0.9774</b> | <b>0.7130</b> |
|     |                       | 80:20 | 0.8776        | 0.5161        | <b>0.9679</b> | 0.7420        | <b>0.8848</b> | <b>0.6386</b> | 0.9464        | <b>0.7925</b> |
|     |                       | 70:30 | <b>0.8643</b> | 0.7544        | 0.9113        | 0.8329        | 0.8640        | <b>0.7505</b> | <b>0.9127</b> | <b>0.8316</b> |
|     | RUS                   | 90:10 | 0.8302        | <b>0.9051</b> | 0.8219        | <b>0.8635</b> | <b>0.8466</b> | 0.8845        | <b>0.8424</b> | 0.8634        |
|     |                       | 80:20 | 0.8379        | <b>0.8912</b> | 0.8246        | 0.8579        | <b>0.8504</b> | 0.8742        | <b>0.8445</b> | <b>0.8593</b> |
|     |                       | 70:30 | 0.8388        | <b>0.8865</b> | 0.8183        | <b>0.8524</b> | <b>0.8472</b> | 0.8602        | <b>0.8416</b> | 0.8509        |
|     | ROS                   | 90:10 | 0.8329        | <b>0.9051</b> | 0.8249        | <b>0.8650</b> | <b>0.8426</b> | 0.8792        | <b>0.8385</b> | 0.8589        |
|     |                       | 80:20 | 0.8371        | <b>0.8886</b> | 0.8242        | 0.8564        | <b>0.8584</b> | 0.8707        | <b>0.8553</b> | <b>0.8630</b> |
|     |                       | 70:30 | 0.8398        | <b>0.8914</b> | 0.8177        | 0.8546        | <b>0.8566</b> | 0.8736        | <b>0.8494</b> | <b>0.8615</b> |
|     | SMOTE                 | 90:10 | 0.8328        | <b>0.9057</b> | 0.8247        | <b>0.8652</b> | <b>0.8529</b> | 0.8743        | <b>0.8505</b> | 0.8624        |
|     |                       | 80:20 | 0.8376        | <b>0.8904</b> | 0.8244        | 0.8574        | <b>0.8509</b> | 0.8686        | <b>0.8464</b> | <b>0.8575</b> |
|     |                       | 70:30 | 0.8443        | <b>0.8733</b> | 0.8318        | <b>0.8526</b> | <b>0.8538</b> | 0.8455        | <b>0.8574</b> | 0.8515        |
|     | ANS                   | 90:10 | 0.8329        | <b>0.9045</b> | 0.8249        | <b>0.8647</b> | <b>0.8543</b> | 0.8455        | <b>0.8581</b> | 0.8518        |
|     |                       | 80:20 | 0.8378        | <b>0.8908</b> | 0.8246        | <b>0.8577</b> | <b>0.8516</b> | 0.8662        | <b>0.8479</b> | 0.8570        |
|     |                       | 70:30 | 0.8443        | <b>0.8733</b> | 0.8318        | <b>0.8526</b> | <b>0.8543</b> | 0.8455        | <b>0.8581</b> | 0.8518        |
|     | DBSMOTE               | 90:10 | 0.8395        | <b>0.8965</b> | 0.8332        | <b>0.8648</b> | <b>0.8747</b> | 0.8311        | <b>0.8795</b> | 0.8553        |
|     |                       | 80:20 | 0.8435        | <b>0.8829</b> | 0.8337        | <b>0.8583</b> | <b>0.8619</b> | 0.8424        | <b>0.8667</b> | 0.8546        |
|     |                       | 70:30 | 0.8467        | <b>0.8676</b> | 0.8378        | <b>0.8527</b> | <b>0.8604</b> | 0.8281        | <b>0.8742</b> | 0.8512        |

**Table I.** (contd.)

| $n$   | Handling<br>Technique | IR    | MLE    |               |               |               | SCORE         |               |               |               |
|-------|-----------------------|-------|--------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|       |                       |       | Acc    | Sens          | Spec          | BA            | Acc           | Sens          | Spec          | BA            |
| 1,000 | Original              | 90:10 | 0.9056 | 0.0616        | 0.9766        | 0.5305        | <b>0.9251</b> | <b>0.4503</b> | <b>0.9779</b> | <b>0.7141</b> |
|       |                       | 80:20 | 0.8787 | 0.5083        | <b>0.9713</b> | 0.7397        | <b>0.8878</b> | <b>0.6390</b> | 0.9501        | <b>0.7945</b> |
|       |                       | 70:30 | 0.8669 | 0.7526        | <b>0.9159</b> | 0.8342        | <b>0.8672</b> | <b>0.7583</b> | 0.9138        | <b>0.8360</b> |
|       | RUS                   | 90:10 | 0.8340 | <b>0.8949</b> | 0.8272        | <b>0.8610</b> | <b>0.8502</b> | 0.8715        | <b>0.8478</b> | 0.8597        |
|       |                       | 80:20 | 0.8391 | <b>0.8932</b> | 0.8256        | <b>0.8594</b> | <b>0.8503</b> | 0.8720        | <b>0.8449</b> | 0.8584        |
|       |                       | 70:30 | 0.8473 | <b>0.8971</b> | 0.8259        | <b>0.8615</b> | <b>0.8537</b> | 0.8732        | <b>0.8453</b> | 0.8593        |
|       | ROS                   | 90:10 | 0.8333 | <b>0.8952</b> | 0.8264        | <b>0.8608</b> | <b>0.8510</b> | 0.8718        | <b>0.8487</b> | 0.8602        |
|       |                       | 80:20 | 0.8388 | <b>0.8923</b> | 0.8254        | 0.8588        | <b>0.8517</b> | 0.8709        | <b>0.8468</b> | <b>0.8589</b> |
|       |                       | 70:30 | 0.8482 | <b>0.8951</b> | 0.8281        | <b>0.8616</b> | <b>0.8556</b> | 0.8733        | <b>0.8481</b> | 0.8607        |
|       | SMOTE                 | 90:10 | 0.8335 | <b>0.8962</b> | 0.8265        | <b>0.8614</b> | <b>0.8523</b> | 0.8699        | <b>0.8503</b> | 0.8601        |
|       |                       | 80:20 | 0.8396 | <b>0.8926</b> | 0.8263        | <b>0.8595</b> | <b>0.8519</b> | 0.8693        | <b>0.8476</b> | 0.8584        |
|       |                       | 70:30 | 0.8523 | <b>0.8780</b> | 0.8413        | <b>0.8597</b> | <b>0.8587</b> | 0.8573        | <b>0.8594</b> | 0.8583        |
|       | ANS                   | 90:10 | 0.8335 | <b>0.8959</b> | 0.8266        | <b>0.8612</b> | <b>0.8537</b> | 0.8671        | <b>0.8522</b> | 0.8596        |
|       |                       | 80:20 | 0.8395 | <b>0.8924</b> | 0.8262        | <b>0.8593</b> | <b>0.8527</b> | 0.8660        | <b>0.8494</b> | 0.8577        |
|       |                       | 70:30 | 0.8528 | <b>0.8789</b> | 0.8416        | <b>0.8603</b> | <b>0.8589</b> | 0.8558        | <b>0.8602</b> | 0.8580        |
|       | DBSMOTE               | 90:10 | 0.8404 | <b>0.8855</b> | 0.8358        | <b>0.8621</b> | <b>0.8720</b> | 0.8309        | <b>0.8766</b> | 0.8537        |
|       |                       | 80:20 | 0.8445 | <b>0.8831</b> | 0.8349        | <b>0.8590</b> | <b>0.8649</b> | 0.8357        | <b>0.8722</b> | 0.8540        |
|       |                       | 70:30 | 0.8533 | <b>0.8711</b> | 0.8457        | <b>0.8584</b> | <b>0.8599</b> | 0.8381        | <b>0.8693</b> | 0.8537        |

**Note:** Bold font indicates the superior result between MLE and SCORE estimators.

**Table II.** Mean predictive performance of the Logistic Regression model with three independent variables

| $n$ | Handling<br>Technique | IR    | MLE    |               |               |               | Score         |               |               |               |
|-----|-----------------------|-------|--------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|     |                       |       | Acc    | Sens          | Spec          | BA            | Acc           | Sens          | Spec          | BA            |
| 200 | Original              | 90:10 | 0.9222 | 0.2232        | 0.9438        | 0.6115        | <b>0.9402</b> | <b>0.5842</b> | <b>0.9798</b> | <b>0.7820</b> |
|     |                       | 80:20 | 0.9027 | 0.6200        | <b>0.9733</b> | 0.7967        | <b>0.9150</b> | <b>0.7549</b> | 0.9550        | <b>0.8550</b> |
|     |                       | 70:30 | 0.8960 | <b>0.8352</b> | 0.9220        | <b>0.8786</b> | <b>0.8965</b> | 0.8224        | <b>0.9283</b> | 0.8754        |
|     | RUS                   | 90:10 | 0.8402 | <b>0.9378</b> | 0.8293        | 0.8836        | <b>0.8679</b> | 0.9262        | <b>0.8614</b> | <b>0.8938</b> |
|     |                       | 80:20 | 0.8592 | <b>0.9465</b> | 0.8373        | 0.8919        | <b>0.8826</b> | 0.9224        | <b>0.8727</b> | <b>0.8976</b> |
|     |                       | 70:30 | 0.8710 | <b>0.9492</b> | 0.8375        | 0.8933        | <b>0.8865</b> | 0.9134        | <b>0.8750</b> | <b>0.8942</b> |
|     | ROS                   | 90:10 | 0.8530 | <b>0.9520</b> | 0.8419        | 0.8970        | <b>0.8946</b> | 0.9217        | <b>0.8916</b> | <b>0.9066</b> |
|     |                       | 80:20 | 0.8677 | <b>0.9493</b> | 0.8473        | 0.8983        | <b>0.8946</b> | 0.9133        | <b>0.8899</b> | <b>0.9016</b> |
|     |                       | 70:30 | 0.8728 | <b>0.9510</b> | 0.8392        | 0.8951        | <b>0.8889</b> | 0.9128        | <b>0.8786</b> | <b>0.8957</b> |
|     | SMOTE                 | 90:10 | 0.8402 | <b>0.9378</b> | 0.8293        | 0.8836        | <b>0.9050</b> | 0.8887        | <b>0.9068</b> | <b>0.8977</b> |
|     |                       | 80:20 | 0.8592 | <b>0.9465</b> | 0.8373        | 0.8919        | <b>0.9002</b> | 0.9070        | <b>0.8985</b> | <b>0.9028</b> |
|     |                       | 70:30 | 0.8710 | <b>0.9492</b> | 0.8375        | <b>0.8933</b> | <b>0.8937</b> | 0.8918        | <b>0.8945</b> | 0.8931        |
|     | ANS                   | 90:10 | 0.8588 | <b>0.9387</b> | 0.8499        | <b>0.8943</b> | <b>0.9089</b> | 0.8713        | <b>0.9131</b> | 0.8922        |
|     |                       | 80:20 | 0.8690 | <b>0.9516</b> | 0.8484        | 0.9000        | <b>0.9004</b> | 0.9013        | <b>0.9002</b> | <b>0.9007</b> |
|     |                       | 70:30 | 0.8793 | <b>0.9429</b> | 0.8521        | <b>0.8975</b> | <b>0.8944</b> | 0.8903        | <b>0.8962</b> | 0.8932        |
|     | DBSMOTE               | 90:10 | 0.8612 | <b>0.9380</b> | 0.8527        | <b>0.8954</b> | <b>0.9123</b> | 0.8645        | <b>0.9176</b> | 0.8911        |
|     |                       | 80:20 | 0.8711 | <b>0.9479</b> | 0.8518        | <b>0.8999</b> | <b>0.9052</b> | 0.8789        | <b>0.9117</b> | 0.8953        |
|     |                       | 70:30 | 0.8807 | <b>0.9384</b> | 0.8560        | <b>0.8972</b> | <b>0.8957</b> | 0.8796        | <b>0.9026</b> | 0.8911        |
| 500 | Original              | 90:10 | 0.9106 | 0.1074        | 0.9613        | 0.5536        | <b>0.9410</b> | <b>0.5997</b> | <b>0.9789</b> | <b>0.7893</b> |
|     |                       | 80:20 | 0.8995 | 0.6059        | 0.9730        | 0.7894        | <b>0.9063</b> | <b>0.7357</b> | <b>0.9490</b> | <b>0.8423</b> |
|     |                       | 70:30 | 0.9009 | <b>0.8372</b> | 0.9281        | <b>0.8827</b> | <b>0.9012</b> | 0.8297        | <b>0.9318</b> | 0.8807        |
|     | RUS                   | 90:10 | 0.8484 | <b>0.9383</b> | 0.8384        | 0.8883        | <b>0.8795</b> | 0.9121        | <b>0.8758</b> | <b>0.8940</b> |
|     |                       | 80:20 | 0.8552 | <b>0.9406</b> | 0.8339        | 0.8872        | <b>0.8786</b> | 0.9034        | <b>0.8724</b> | <b>0.8879</b> |
|     |                       | 70:30 | 0.8791 | <b>0.9471</b> | 0.8499        | <b>0.8985</b> | <b>0.8926</b> | 0.9047        | <b>0.8874</b> | 0.8960        |
|     | ROS                   | 90:10 | 0.8532 | <b>0.9445</b> | 0.8430        | 0.8938        | <b>0.8890</b> | 0.9073        | <b>0.8869</b> | <b>0.8971</b> |
|     |                       | 80:20 | 0.8574 | <b>0.9425</b> | 0.8361        | 0.8893        | <b>0.8847</b> | 0.9072        | <b>0.8790</b> | <b>0.8931</b> |
|     |                       | 70:30 | 0.8762 | <b>0.9474</b> | 0.8457        | 0.8966        | <b>0.8914</b> | 0.9069        | <b>0.8847</b> | <b>0.8958</b> |
|     | SMOTE                 | 90:10 | 0.8524 | <b>0.9449</b> | 0.8422        | 0.8935        | <b>0.8977</b> | 0.8913        | <b>0.8984</b> | <b>0.8949</b> |
|     |                       | 80:20 | 0.8593 | <b>0.9456</b> | 0.8378        | <b>0.8917</b> | <b>0.8887</b> | 0.8934        | <b>0.8876</b> | 0.8905        |
|     |                       | 70:30 | 0.8824 | <b>0.9386</b> | 0.8584        | <b>0.8985</b> | <b>0.8976</b> | 0.8894        | <b>0.9010</b> | 0.8952        |
|     | ANS                   | 90:10 | 0.8557 | <b>0.9420</b> | 0.8461        | 0.8940        | <b>0.9014</b> | 0.8852        | <b>0.9032</b> | <b>0.8942</b> |
|     |                       | 80:20 | 0.8593 | <b>0.9453</b> | 0.8378        | <b>0.8915</b> | <b>0.8899</b> | 0.8882        | <b>0.8903</b> | 0.8893        |
|     |                       | 70:30 | 0.8822 | <b>0.9381</b> | 0.8582        | <b>0.8982</b> | <b>0.8982</b> | 0.8863        | <b>0.9034</b> | 0.8948        |
|     | DBSMOTE               | 90:10 | 0.8568 | <b>0.9360</b> | 0.8480        | <b>0.8920</b> | <b>0.9116</b> | 0.8563        | <b>0.9178</b> | 0.8870        |
|     |                       | 80:20 | 0.8626 | <b>0.9375</b> | 0.8439        | <b>0.8907</b> | <b>0.8968</b> | 0.8579        | <b>0.9065</b> | 0.8822        |
|     |                       | 70:30 | 0.8824 | <b>0.9359</b> | 0.8595        | <b>0.8977</b> | <b>0.9009</b> | 0.8694        | <b>0.9144</b> | 0.8919        |

**Table II.** (contd.)

| $n$   | Handling<br>Technique | IR    | MLE    |               |               |               | Score         |               |               |               |
|-------|-----------------------|-------|--------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|       |                       |       | Acc    | Sens          | Spec          | BA            | Acc           | Sens          | Spec          | BA            |
| 1,000 | Original              | 90:10 | 0.9070 | 0.0703        | 0.9751        | 0.5351        | <b>0.9366</b> | <b>0.5663</b> | <b>0.9778</b> | <b>0.7720</b> |
|       |                       | 80:20 | 0.8970 | 0.5684        | <b>0.9791</b> | 0.7738        | <b>0.9037</b> | <b>0.7095</b> | 0.9522        | <b>0.8309</b> |
|       |                       | 70:30 | 0.8978 | <b>0.8370</b> | 0.9238        | <b>0.8804</b> | <b>0.9063</b> | 0.7357        | <b>0.9490</b> | 0.8423        |
|       | RUS                   | 90:10 | 0.8479 | <b>0.9557</b> | 0.8359        | 0.8958        | <b>0.8824</b> | 0.9134        | <b>0.8790</b> | <b>0.8962</b> |
|       |                       | 80:20 | 0.8569 | <b>0.9301</b> | 0.8386        | <b>0.8843</b> | <b>0.8761</b> | 0.8844        | <b>0.8741</b> | 0.8793        |
|       |                       | 70:30 | 0.8715 | <b>0.9469</b> | 0.8391        | 0.8930        | <b>0.8882</b> | 0.9110        | <b>0.8784</b> | <b>0.8947</b> |
|       | ROS                   | 90:10 | 0.8518 | <b>0.9539</b> | 0.8405        | <b>0.8972</b> | <b>0.8838</b> | 0.9134        | <b>0.8805</b> | 0.8970        |
|       |                       | 80:20 | 0.8569 | <b>0.9422</b> | 0.8356        | <b>0.8889</b> | <b>0.8771</b> | 0.8966        | <b>0.8722</b> | 0.8844        |
|       |                       | 70:30 | 0.8725 | <b>0.9465</b> | 0.8407        | 0.8936        | <b>0.8913</b> | 0.9107        | <b>0.8830</b> | <b>0.8969</b> |
|       | SMOTE                 | 90:10 | 0.8495 | <b>0.9578</b> | 0.8375        | <b>0.8976</b> | <b>0.8920</b> | 0.9044        | <b>0.8906</b> | 0.8975        |
|       |                       | 80:20 | 0.8545 | <b>0.9301</b> | 0.8356        | 0.8828        | <b>0.8837</b> | 0.8883        | <b>0.8825</b> | <b>0.8854</b> |
|       |                       | 70:30 | 0.8782 | <b>0.9362</b> | 0.8533        | 0.8948        | <b>0.8945</b> | 0.8977        | <b>0.8931</b> | <b>0.8954</b> |
|       | ANS                   | 90:10 | 0.8506 | <b>0.9557</b> | 0.8390        | <b>0.8973</b> | <b>0.8947</b> | 0.8993        | <b>0.8942</b> | 0.8968        |
|       |                       | 80:20 | 0.8537 | <b>0.9301</b> | 0.8346        | 0.8823        | <b>0.8829</b> | 0.8880        | <b>0.8816</b> | <b>0.8848</b> |
|       |                       | 70:30 | 0.8785 | <b>0.9364</b> | 0.8537        | <b>0.8950</b> | <b>0.8950</b> | 0.8939        | <b>0.8955</b> | 0.8947        |
|       | DBSMOTE               | 90:10 | 0.8508 | <b>0.9484</b> | 0.8399        | <b>0.8942</b> | <b>0.9081</b> | 0.8675        | <b>0.9126</b> | 0.8901        |
|       |                       | 80:20 | 0.8602 | <b>0.9380</b> | 0.8408        | <b>0.8894</b> | <b>0.8879</b> | 0.8588        | <b>0.8952</b> | 0.8770        |
|       |                       | 70:30 | 0.8789 | <b>0.9339</b> | 0.8553        | <b>0.8946</b> | <b>0.8977</b> | 0.8806        | <b>0.9051</b> | 0.8928        |

**Note:** Bold font indicates the superior result between MLE and SCORE estimators.